

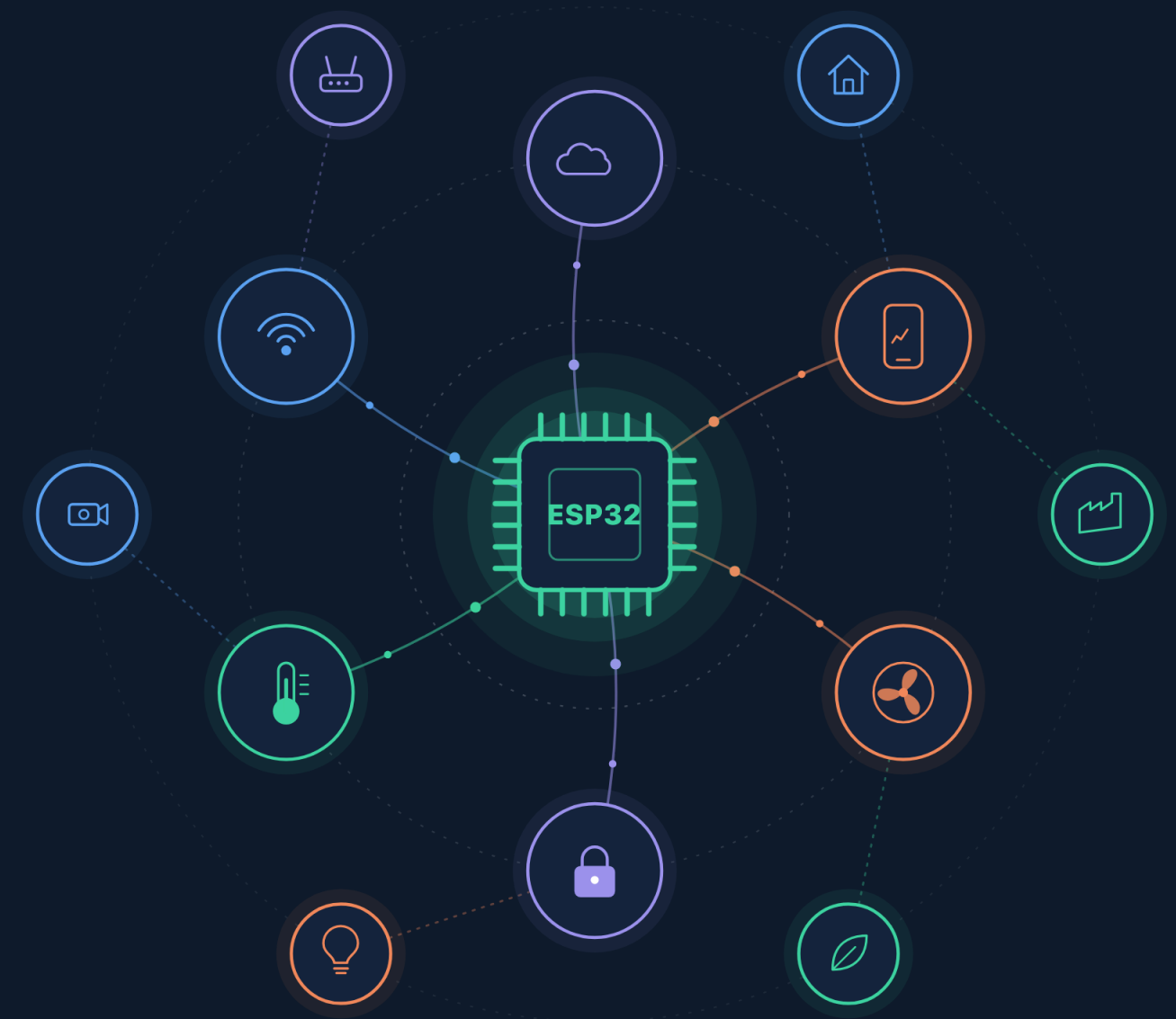
MODULE 4 • 45 MINUTES

Getting Online: Connectivity and Communication Protocols

How devices connect, the “languages” they speak once connected, and how to put your own device online.

Mehdi Lotfinejad

Introduction to the Internet of Things



THE ROAD AHEAD

Eight modules, one smart device

PART 1 — DEVICES AND DATA

01

What is IoT?

Lecture

02

How it's built

Lecture

03

Sensors and actuators

Labs 1–2

04

Getting online

You are here

PART 2 — CLOUD, AUTOMATION AND SECURITY

05

Cloud and dashboards

Lab 4

06

Automation

Lab 5

07

Security

Group activity

08

Real world + capstone

Lab 6

BY THE END OF THIS MODULE

You will be able to...

-
- 01 **Explain** addresses, clients and servers, ports and packets
 - 02 **Compare** Wi-Fi, BLE, Zigbee, LoRaWAN and cellular IoT
 - 03 **Choose** a suitable connection for an IoT project
 - 04 **Explain** how HTTP and MQTT work, and when to use each
 - 05 **Connect** a simulated ESP32 and publish live data over MQTT
-

Two separate decisions



Protocol

The **language**: MQTT, HTTP/REST, CoAP



Network

Addressing and routing: IP addresses, the internet



Connection

The **road**: Wi-Fi, BLE, Zigbee, LoRaWAN, cellular, Ethernet

In Lab 3: **Wi-Fi** is the road and **MQTT** is the language. In Lab 4: still Wi-Fi, but **HTTP**.

Part A

Networking basics

A handful of words you will keep meeting. You don't need to be a network engineer.

Addresses, names and servers



IP address

Every device's address, like 192.168.1.42



DNS

The phone book: turns names into IP addresses



Router / gateway

Passes data between networks, or radio types



Client / server

The client asks; the server listens and answers

Ports, packets and speed



Port

Which service: 80, 443, MQTT
1883



Packets

Data travels in small chunks



TCP

Reliable, in order: registered post



UDP

Lighter, no promises: ordinary post



Bandwidth

Data per second: the road's width



Latency

How long one message takes to
arrive

PAUSE AND CHECK

Your ESP32 needs to send a reading to `broker.hivemq.com`. What turns that name into an address the network can use?

Part B

Choosing the road

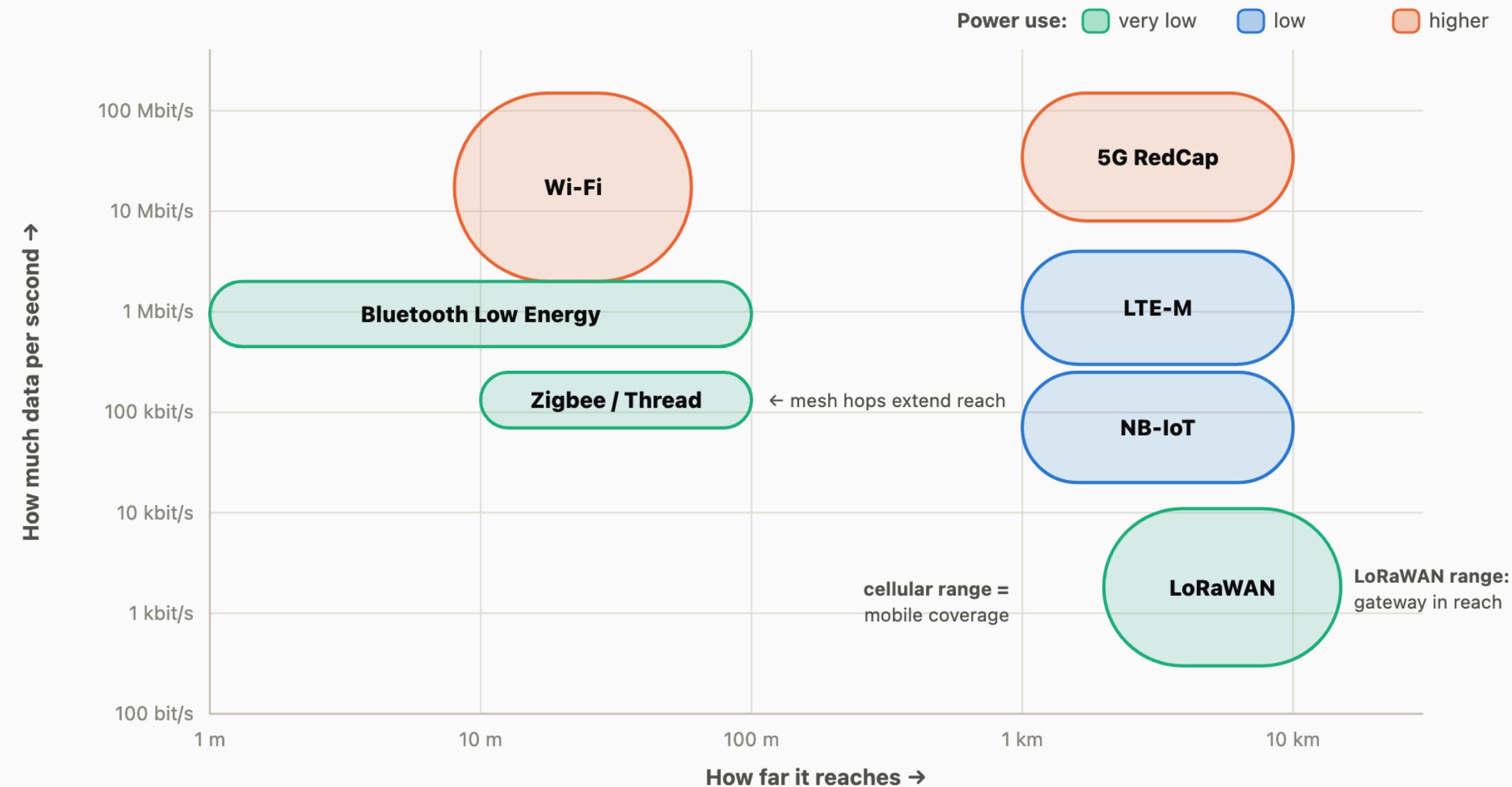
Wi-Fi, Bluetooth, Zigbee, LoRaWAN, mobile networks — and how to pick one.

THE BIG TRADE-OFF

Range, power and data rate

Range vs data rate — and what it costs in power

Typical values, log scales. Colour = power use: the only options with long range AND lots of data use more power.



Wi-Fi: lots of data, but high power

BLE: very low power, short range

LoRaWAN: kilometres, but tiny data

SHORT RANGE

Wi-Fi, Bluetooth LE and Zigbee



Wi-Fi

~30–50 m indoors. Lots of data, high power. Plugged-in devices, cameras.



Bluetooth LE

~10 m. Very low power, years on a coin battery. Wearables, locks.



Zigbee / Thread

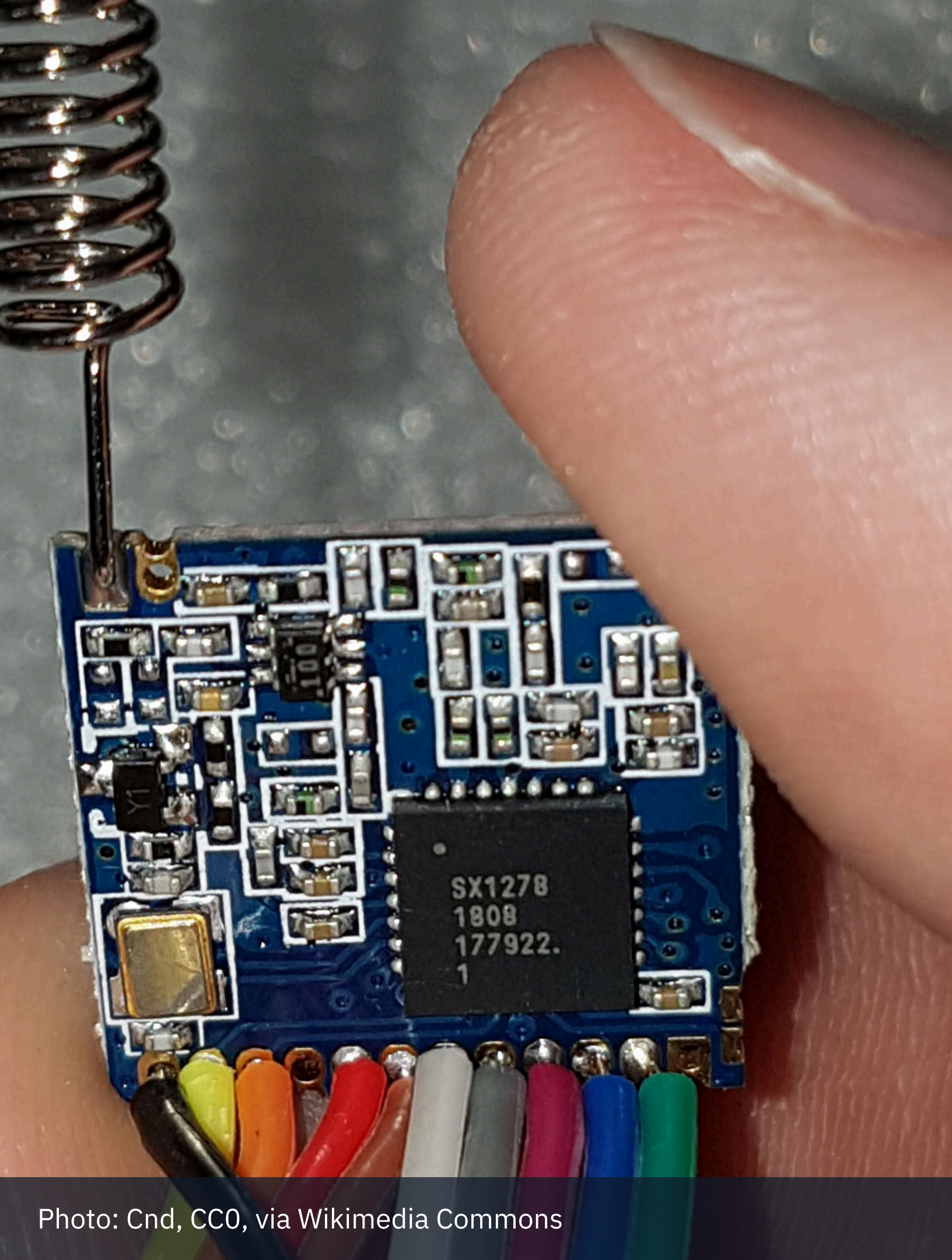
A **mesh**: devices pass messages on. Needs a hub. Smart lights.



LONG RANGE

LoRaWAN

- Often **over 10 km** in the countryside
- A few short messages an hour
- Up to **10 years** on a battery
- Needs a **gateway** within range



INSIDE A LORA SENSOR

A small radio, the right band

- No licence needed for the radio band
- The band differs by country
- Malaysia: **AS923**, about 919–924 MHz

SIDE BY SIDE

Comparing the options

Technology	Range	Power	Needs	Typical use
Wi-Fi	~30–50 m indoors	High	A Wi-Fi router	Cameras, smart plugs
Bluetooth LE	~10 m+	Very low	A phone or gateway	Wearables, locks
Zigbee / Thread	Short, mesh extends	Very low	A hub	Lights, switches
LoRaWAN	Kilometres	Extremely low	A LoRaWAN gateway	Farms, cities
NB-IoT / LTE-M	Mobile coverage	Low	Nothing extra	Meters, trackers
Ethernet	Cable length	Mains	A network switch	Factories

Typical values only. Malaysia switched off 3G in 2022: new cellular products use **NB-IoT, LTE-M or 5G**.

PAUSE AND CHECK

Pick a technology: (a) a river sensor 8 km from town, on a battery; (b) an office security camera; (c) a heart-rate strap that talks to a phone.

Ask: How far? Mains or battery? How much data? What already exists there?

Part C

The languages devices speak

HTTP, MQTT and CoAP — and how to keep the messages safe.

Four methods, a few status codes

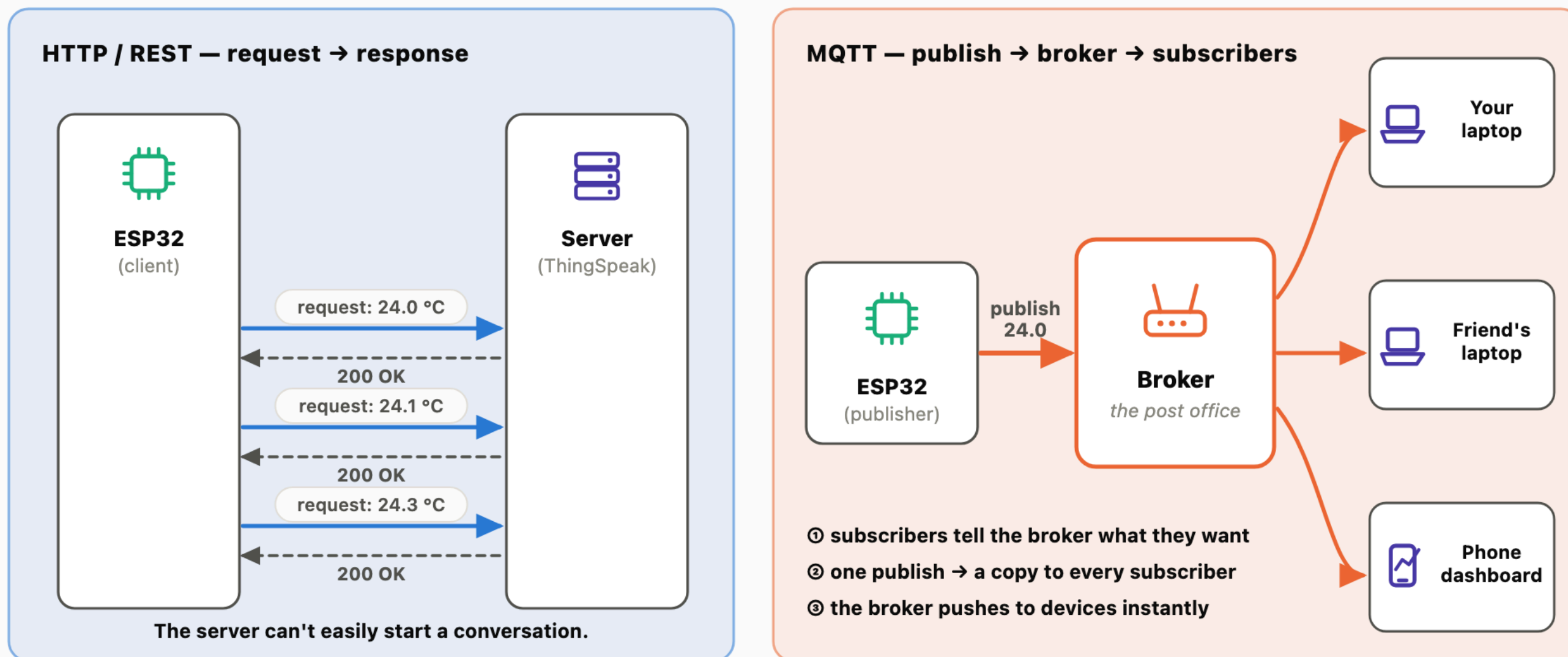
Method	Means	Example
GET	Read something	Get the latest reading
POST	Create or send something new	Send a new reading
PUT	Update something	Change a device’s settings
DELETE	Remove something	Delete a device

Status codes: **200** success · **400** your request was wrong · **404** not found · **500** server problem.

HTTP asks; MQTT posts through a broker

Two ways to talk: ask-and-answer vs the post office

HTTP: the device asks, the server answers — every time. MQTT: publish once, the broker delivers to every subscriber.



Topics and wildcards

Which messages does a subscription receive?

Topics are levels separated by "/". + matches exactly one level. # matches everything below.

Message published to topic...	subscribed to: iot-workshop/+/temperature	subscribed to: iot-workshop/#	subscribed to: iot-workshop/aisha/#
iot-workshop / aisha / temperature	✓	✓	✓
iot-workshop / aisha / humidity	✗	✓	✓
iot-workshop / ravi / temperature	✓	✓	✗
iot-workshop / ravi / humidity	✗	✓	✗
iot-workshop / ravi / room1 / temperature	✗	✓	✗

two levels where + stands → no

CHOOSING A PROTOCOL

HTTP, MQTT or CoAP?

	HTTP/REST	MQTT	CoAP
Pattern	Request–response	Publish–subscribe	Request–response
Runs on	TCP	TCP	UDP
Message size	Larger	Very small	Very small
Server can push?	Not easily	Yes, instantly	Yes, with “observe”
Best for	Occasional uploads	Frequent, live messages	Tiny battery devices
Our labs	Lab 4	Labs 3 and 6	Not used

Rule of thumb: **HTTP** when a device occasionally reports to a web service; **MQTT** for frequent, live messages.

Four risks, four defences



Eavesdropping

Defence: **encryption**
(TLS, HTTPS, MQTT
on port 8883)



Impersonation

Defence:
authentication for
each device



Open topics

Defence: control
who may publish or
subscribe



Leaked secrets

Passwords and API
keys in shared code
can leak

Lab 3 uses a **public** broker on unencrypted port 1883 with no password. For learning only!

REAL-WORLD EXAMPLE

200,000 smart water meters



Connection

NB-IoT or **LoRaWAN**:
basements, 10+ years on a
battery



Protocol

One tiny message a day:
MQTT, CoAP or a compact
format



Security

Encrypted readings; every
meter has its own identity

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ Myth *Wi-Fi is always best because it's fastest.*

✓ Fact Wi-Fi drains batteries. In a field, LoRaWAN is far better.

✗ Myth *Bluetooth and Wi-Fi do the same job.*

✓ Fact BLE: low power, short range. Wi-Fi: more data, direct internet.

✗ Myth *MQTT devices send messages to each other.*

✓ Fact Every message goes through the broker.

✗ Myth *Data on the internet is secure by default.*

✓ Fact Only with encryption and authentication.

LAB 3 • ABOUT 30 MINUTES

Getting online: Wi-Fi + MQTT



- 1 Open a new ESP32 project in Wokwi
- 2 Paste sketch.ino — change “changeme” to your name
- 3 Paste diagram.json and add libraries.txt
- 4 Press Play: Wi-Fi, then MQTT, then readings
- 5 Watch it live in the HiveMQ web client



Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- ✓ Two decisions: **how to connect** (the road) and **what language** to speak
 - ✓ Every radio trades off **range, power and data rate**
 - ✓ Wi-Fi: lots of data · BLE: low power · LoRaWAN: km, tiny data
 - ✓ **HTTP**: ask and answer · **MQTT**: publish–subscribe via a broker
 - ✓ Real systems need **encryption** and **authentication**
-

Quick quiz

Q1 What is the difference between a connection technology and a protocol?

Q2 Farm soil sensors over 5 km, on batteries for years. Which technology — and why not Wi-Fi?

Q3 Does `iot-workshop/+/humidity` match `iot-workshop/lee/room1/humidity`?

Q4 Why is a public broker on port 1883 risky for a real product?

UP NEXT • MODULE 5

IoT Cloud Platforms and Data Visualisation

We give the data a home: what IoT cloud platforms do, and a live graph of your readings in Lab 4 — using HTTP this time.



Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop